12-18 第2篇

阅读理解（说明文） How Does a Simple Compass Find North?

① On a cold December afternoon, if you place a small round magnetic compass on a flat library table in your school, close your eyes, spin the compass gently three times, and then open your eyes, **the red end of the tiny needle will always — after a few quiet seconds of trembling — settle, pointing in the same direction**. You have not plugged it in. You have not opened a map app. You have not connected to any satellite. And yet the needle, a small piece of magnetised metal no bigger than a grain of rice, knows, quietly, exactly where north is. How? The answer turns out to be one of the most beautiful small facts of our planet: **you are standing, right now, inside an enormous invisible magnet**, and the compass is simply a tiny polite visitor who nods to it.

② **Finding one: Earth itself is a giant magnet.** Deep inside the Earth, at a depth between about **2,900 km and 5,150 km**, there is a layer called the **outer core** — an ocean of molten iron and nickel, about **2,300 km thick**, at temperatures of roughly **4,000 °C to 5,000 °C**. This hot molten metal is not sitting still. It moves in slow, vast, spiral currents, driven by the rotation of the Earth and the small temperature difference between the solid inner core and the cooler mantle above. Moving liquid metal is, in the language of physics, an **electric current** — and every electric current, as the 19th-century Danish scientist **Hans Christian Ørsted** discovered in 1820, creates a magnetic field around it. The result of billions of such moving spirals of iron, over the whole outer core, is a single enormous **planetary magnetic field** that stretches from the centre of the Earth outward, past the surface, past the Moon's orbit, to about **65,000 km into space** on the side facing the Sun (and much further on the opposite side). We live inside this field. We cannot see it, feel it, or taste it. But a small compass needle can feel it perfectly.

③ **Finding two: a compass needle is just a small, free-to-turn magnet.** A compass needle is not magic. It is a tiny strip of steel (or similar metal) that has been **magnetised** in a factory — meaning its own tiny iron atoms have been lined up so that their small internal magnetic fields all point, together, in the same direction along the needle. One end of the needle therefore has a magnetic **north pole**, and the other end has a magnetic **south pole**. The needle is then placed on a sharp thin pivot (a small upright pin, usually made of brass), with a tiny drop of oil or a jewel bearing so it can rotate very freely. Now a simple physics rule takes over: **any free-to-turn magnet, placed inside an outside magnetic field, will slowly rotate until its own north pole lines up with the outside field.** Earth's field points (roughly) from the **Earth's geographic south** toward the **Earth's geographic north**. So the north end of any compass needle, after spinning, settles pointing (roughly) toward geographic north. That is the whole story.

④ **Why "roughly" north — not exactly north?** Because of one small beautiful complication. The **geographic North Pole** — the top of the Earth's spinning axis — is **not** the same point as the **magnetic North Pole** — the place where Earth's magnetic field actually points straight down into the ground. Currently (2024 measurements, World Magnetic Model, published by the British Geological Survey and NOAA), the magnetic North Pole sits in the **Arctic Ocean, around 86° N, 139° W**, about **400 km** from the geographic North Pole, and — this is the beautiful part — **it moves**. It has moved from northern Canada in 1900, across the Arctic, toward Siberia, at a current speed of about **55 km per year**. The difference between where your compass points and true geographic north is called the **magnetic declination**. In Beijing today (2025), declination is about **−6°** (the compass points 6° west of true north). Careful hikers, sailors, and pilots learn to correct for this.

⑤ **Finding three: the Earth's magnetic poles have flipped many times.** Deep in the rocks of the Atlantic Ocean floor, when scientists in the 1960s measured the direction of small fossilised magnetism in the lava stripes, they found that the Earth's magnetic north and south have **swapped places** many times — at least **184 times** over the last **83 million years**. The last full reversal, called the **Brunhes–Matuyama reversal**, happened about **780,000 years ago**. During such a flip, which takes roughly **1,000 to 10,000 years** to complete, the magnetic field does not simply turn upside down instantly; it weakens, becomes briefly multipolar, and then reorganises around a new stable north. A school compass during a future flip will, slowly, start pointing in a very different direction than today. There is no reason to worry — such flips have happened many times during human history (well, pre-human history) without harming life.

⑥ **What disturbs a compass?** A few small quiet things.

Disturber	How it affects the compass	How to avoid
Iron and steel objects (radiators, metal chairs, your pencil-case clip)	Local metal has its own small magnetic moments that distort the nearby field	Hold the compass away from large metal objects, at least 50 cm
Electric wires carrying current (in walls, behind a TV)	Moving charge creates a local magnetic field that can shift the needle	Avoid using a compass right against a wall socket or a laptop
Strong permanent magnets (fridge magnets, some earphones, some iPad covers with magnetic clasps)	A close strong magnet completely overpowers Earth's gentle field	Keep such magnets more than 30 cm away
Solar storms (large bursts from the Sun)	Waves of charged particles briefly distort the upper parts of Earth's field	Rare; barely noticeable on small school compasses
Magnetic rocks (iron-rich mountains)	Local stones with strong magnetisation can pull the needle	Mountaineers learn to cross-check with a second compass

⑦ **A 3,000-year-old idea.** The compass, quietly, is not a recent invention. Around the 2nd century BCE, **Chinese scholars in the Han dynasty** first discovered that a natural stone called **lodestone** (a naturally magnetised lump of magnetite) would swing, when suspended, toward the south. By the **11th century (Song dynasty)**, Chinese sailors were using small iron-needle compasses floating on water bowls to cross the South China Sea at night — a full two centuries before Europeans adopted the same tool. The needle you spin today in a school library, therefore, is a direct quiet descendant of a Chinese fisherman's bowl of water, one thousand years ago.

⑧ **Why does the compass still matter in the age of GPS?** A GPS phone tells you, with great accuracy, **where** you are — but it does not tell you **which way you are facing**, unless you start walking. A compass tells you which way you are facing while you are standing still, without batteries, without a satellite, without electricity. In a power cut, in a deep forest, in a boat adrift, on a winter mountain when the phone is cold-dead, the small needle still quietly points. The **Himalayan mountain-rescue services** (Nepal, 2022 data) still require **two non-electronic compasses per rescue team**, by protocol. GPS is newer; the compass is older; both, wisely used, keep humans unlost.

⑨ **The quiet lesson.** Put a cheap compass on your winter desk. Watch the red end tremble, search, and settle. You are looking at one of the shortest, cleanest demonstrations of deep physics available in any living room: a small lined-up magnet rotating, in perfect silence, inside the enormous invisible magnetic field of the planet below you. No electricity. No battery. No code. Just a small piece of metal, quietly nodding to the four thousand-degree ocean of iron at the centre of the Earth — and telling you, kindly, which way is home.

注释： compass 指南针；outer core 外核；mantle 地幔；magnetise 磁化；pivot 枢轴；jewel bearing 宝石轴承；declination 偏角；lodestone 磁石；multipolar 多极的

根据短文内容，选择最佳答案。

1. According to Paragraph ②, what is the physical cause of Earth's magnetic field?

A. A giant magnetic bar buried at the top of the North Pole.
B. Deep inside the Earth, at a depth between about 2,900 km and 5,150 km, lies a 2,300 km-thick outer core of molten iron and nickel at 4,000—5,000 °C; this hot molten metal moves in slow, vast, spiral currents driven by the Earth's rotation and the temperature difference between the solid inner core and the cooler mantle above; because moving liquid metal is an electric current, and every electric current (as Hans Christian Ørsted discovered in 1820) creates a magnetic field, billions of such spiral currents across the whole outer core together produce a single enormous planetary magnetic field that reaches about 65,000 km into space on the Sun-facing side — we live inside this field.
C. The Sun's rays reflect off the ground and magnetise the Earth's surface.
D. Every person's body produces a magnetic field that adds together.

2. According to Paragraph ③, how exactly does a compass needle work?

A. A compass needle is magical and needs no explanation.
B. A compass needle is a tiny strip of steel magnetised in a factory — its own tiny iron atoms are lined up so that their small internal magnetic fields all point together in the same direction along the needle, giving one end a magnetic north pole and the other a south pole; the needle is placed on a sharp thin pivot (usually a small upright brass pin, with a drop of oil or a jewel bearing) so it can rotate very freely; a simple physics rule then takes over: any free-to-turn magnet placed inside an outside magnetic field will slowly rotate until its own north pole lines up with the outside field, and because Earth's field points roughly from geographic south toward geographic north, the north end of any compass needle settles pointing roughly toward geographic north.
C. A compass needle runs on tiny hidden batteries that power an internal motor.
D. A compass needle points at the largest object in the room.

3. What is "magnetic declination", and why does a compass not point EXACTLY to geographic north (Paragraph ④)?

A. Magnetic declination is a weather phenomenon.
B. The magnetic North Pole is exactly on the geographic North Pole.
C. The geographic North Pole (the top of Earth's spinning axis) is not the same as the magnetic North Pole (where Earth's field points straight down into the ground); 2024 data from the World Magnetic Model (BGS and NOAA) put the magnetic North Pole at about 86° N, 139° W in the Arctic Ocean, around 400 km from the geographic pole, and it has moved from northern Canada in 1900 toward Siberia at about 55 km per year; the angle between where the compass points and true geographic north is called magnetic declination (in Beijing today this is about −6°, meaning the compass points about 6° west of true north), and careful hikers, sailors and pilots learn to correct for it.
D. The compass points at the magnetic South Pole instead.

4. What can disturb a school compass, and what is the quiet rule about using it safely (Paragraph ⑥)?

A. Only sunlight and water disturb a compass.
B. Pencils and paper strongly disturb a compass.
C. Heavy boots always ruin the compass permanently.
D. Iron and steel objects (radiators, metal chairs, pencil-case clips) distort the nearby field and should be kept at least 50 cm away; electric wires carrying current (in walls, behind TVs) shift the needle so students should avoid a compass right against a wall socket or laptop; strong permanent magnets (fridge magnets, some earphones, magnetic iPad covers) can completely overpower Earth's gentle field so they must be kept more than 30 cm away; solar storms briefly distort the upper field but are barely noticeable on small school compasses; and iron-rich magnetic rocks in mountains can locally pull the needle, so mountaineers learn to cross-check with a second compass.

5. What is the main lesson of the final paragraph?

A. Compasses are useless today and should be thrown away.
B. GPS has completely replaced the compass in every use case.
C. A cheap compass on a winter desk, with its red end trembling, searching and settling, is one of the shortest and cleanest demonstrations of deep physics available in any living room — a small lined-up magnet rotating in perfect silence inside the planet's enormous invisible magnetic field, with no electricity, no battery and no code, quietly nodding to the four-thousand-degree ocean of iron at the centre of the Earth and telling us, kindly, which way is home.
D. Lodestones are fake rocks.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段地球磁场成因
【解析】"外核 / 2300 km / 4000–5000 ℃ / 液态金属电流 / Ørsted 1820 / 65000 km 延伸"直引。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段指针原理
【解析】"磁化 / 两极 / 枢轴 / 自由磁铁转向外磁场"直引。A、C、D 均错。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第④段偏角
【解析】"2024 / 86°N 139°W / 400 km / 55 km/年 / 北京 −6°"直引。A、B、D 均错。故选 C。

4.
【答案】D
【题型】细节理解题
【定位】第⑥段干扰表
【解析】"铁 50 cm / 电线插座 / 强磁 30 cm / 太阳风 / 磁性岩石"直引。A、B、C 均错。故选 D。

5.
【答案】C
【题型】主旨大意题

【定位】第⑨段结尾
【解析】"最短物理演示 / 无电无电池无代码 / 向 4000 ℃ 铁海低头 / 指向家"主旨。A、B、D 均偏离。故选 C。